

Conclusion: Summary, Contributions, and Synthesis

Summary of Principal Contributions

This paper has developed a unified category-theoretic framework for analyzing linguistic case systems, synthesizing five distinct research traditions into a coherent whole and embedding it within an active inference model of cognition. Our eleven principal contributions are:

C1: Case Categories as a Formal Algebraic Framework

The review formalized case systems as categories with case roles as objects and grammatical relations as morphisms (??). This formalization captures the full typological range of alignment systems—nominative-accusative, ergative-absolutive, active-stative, tripartite, and fluid-S—within a single algebraic framework, with alignment functors providing structure-preserving mappings between systems. By tracing the lineage from Fillmore's [-@fillmore1968case] deep cases through Jakobson's binary distinctive features and Dowty's [-@dowty1991thematic] proto-roles, the analysis demonstrated how enriched (weighted) morphisms link the categorical formalization to the gradient nature of thematic role assignment.

Future Research Directions

F1: Computational Experiments with DisCoCirc and lambeq

The DisCoCirc framework [defelice2020discourse; defelice2022discocirc] offers a natural platform for testing our case-theoretic predictions computationally. The release of **lambeq Gen II** [lambeq2025genii] with full DisCoCirc support makes this direction immediately tractable: discourse-level case role tracking—including the dynamic role reversals of ??—can now be compiled into parameterized quantum circuits and trained end-to-end. Concrete experiments could include:

- ▶ Implementing case-marked DisCoCat models in **lambeq Gen II** and evaluating them on semantic role labeling (SRL) tasks, leveraging the discourse-level wiring to track case roles across sentence boundaries
- ▶ Comparing the accuracy of alignment-specific meaning functors (accusative vs. ergative) on typologically diverse corpora
- ▶ Measuring the categorical magnitude of empirically derived enriched case categories and correlating it with typological

Closing Remarks

The commutative diagram is the central motif of this review—both as a mathematical tool and as a cognitive instrument. The analysis has demonstrated that the same diagrammatic language that makes category theory effective for formalizing case systems also makes it effective for *thinking about* case systems: the spatial structure of a commutative diagram encodes relational information in a format that supports rapid search, pattern recognition, and free-ride inference.

This convergence of mathematical utility and cognitive efficacy is not accidental. If the active inference framework is correct, then the brain operates by constructing and updating generative models of the world's relational structure. Category theory provides the structural algebra for these generative models; commutative diagrams supply their natural topology; and case categories instantiate the precise relational vocabulary that cognitive systems deploy to organize the entropy of experience into coherent narratives of *who does what to whom*. The distributional revolution in both semantics and reinforcement learning—from